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Exploring the impact of individualized pleasure-oriented exercise sessions in a health club setting: Protocol for a randomized controlled trial

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ABSTRACT

Research on the relationship between exercise-induced affect and exercise or physical activity behavior has gained momentum in recent years, yielding several observational and longitudinal studies. However, experimental tests demonstrating a causal role of affective responses on exercise adherence are lacking. Given the need to devise exercise prescriptions that can facilitate adherence and promote regular physical activity, a Randomized Controlled Trial targeting individual pleasurable responses in a health-club setting will be conducted. The experimental protocol will compare two individualized evaluation, prescription, and supervision methods, adjusted for intensity, with the aim to explore their impact on behavioral, affective, and motivational outcomes. The planned study will be a pragmatic randomized, single-blinded, controlled superiority trial with two parallel groups and an allocation ratio of 1:1. Apparently healthy volunteer participants ($N = 46$) between 18 and 45 years old, who are not regularly active, will be randomly allocated to a control or experimental group. The control group will follow a general American College of Sports Medicine exercise prescription; the experimental group will follow the same prescription but with enhancements aimed at promoting pleasure: (1) individualization based on individual differences in preference for and tolerance of exercise intensity; (2) self-regulation of intensity; and (3) repeated assessments of core affect. The primary outcome will be post-intervention exercise-session attendance. Affective and motivational variables will be examined as secondary outcomes. The results are expected to advance exercise prescriptions by promoting affective regulation, thus helping to improve exercise adherence and related outcomes.
Trial registration: ClinicalTrials.gov; ID: NCT05416593.

1. Background and rationale

From a public health standpoint, gymnasia and health clubs are popular settings for supervised exercise practice, attracting millions of individuals worldwide (International Health, Racquet & Sports Club Association, 2020; European Commission, 2022). However, exercise adherence has persistently proven to be a challenge, and several indicators show high attrition rates in these contexts (i.e., dropout in a given period), particularly within the first six months of practice (Buckworth et al., 2013; Sperandei, Vieira, & Reis, 2016).

Individual physical activity promotion can be challenging as it relies on the complex interaction of several processes underlying human behavior. Many psychological theories and strategies have been used to address this issue, albeit with varied results (Rhodes et al., 2019). These are usually based on cognitivist assumptions that have shown small-to-moderate effects on exercise adherence (Ekkekakis & Zenko, 2016; Rhodes & Kates, 2015). However, in recent years, exercise psychology researchers started to shift their attention to other constructs that can help expand the predictive value of current theoretical models. Particularly, affective processes (e.g., emotions, mood, core affect) have been highlighted as relevant when trying to understand or predict behavior, and a call for a new era of "affectivism" is emerging (Dukes et al., 2021). This reflects a transition to a new paradigm, following decades of research during which affective constructs were seen as outcomes, and a relatively short period during which they have been considered as possible predictors and determinants of behavior that can

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expand our understanding of the mechanisms driving human behavior.

In the exercise field, for example, the latest edition of the American College of Sports Medicine (ACSM) guidelines presented an expanded chapter addressing behavioral theories for increasing physical activity. In this chapter, besides the commonly cited theories (e.g., self-efficacy, self-determination theory, theory of planned behavior), the concept of "affect regulation" has been introduced. This is conceptualized as a product of non-conscious motivational processes (e.g., within so-called "dual-process" theories) and automatic associations between repeated episodes of a given behavior (e.g., exercise) and affective responses (ACSM, 2021). This evolution highlights the attempt to approximate the affective processes recent evidence on the study of exercisers' motivation and adherence outcomes.

1.1. Affective determinants in exercise and the role of exercise intensity

Affect can be understood as an umbrella term that encompasses (1) the most general valenced experiential responses (e.g., pleasure/displeasure; good/bad), termed basic affect or *core affect*, and (2) emotion and mood, which reflect underlying appraisal processes, and usually manifest as *distinct affective states* (Ekkekakis & Petruzzello, 2000; Williams et al., 2008) (for a review see Ekkekakis, 2013). In recent years, several theories and models have incorporated affect-related constructs as possible drivers of behavior, including the Affective-Reflective Theory of physical inactivity and exercise (ART; Brand & Ekkekakis, 2018); the Physical Activity Adoption and Maintenance model (PAAM; Strobach et al., 2020); the Theory of Effort Minimization in Physical Activity (TEMPA; Cheval & Boisgontier, 2021); and the Affect and Health Behavior Framework (AHBF; Stevens et al., 2020; Williams & Evans, 2014).

According to the AHBF, the core affective response (i.e., how one feels during or immediately after completing the activity), via automatic or reflective affect processing, can influence motivation, goals, behavioral intentions, and, ultimately, exercise behavior. In this regard, the core affective valence response during exercise has been shown to be a significant predictor of subsequent physical activity behavior (Kwan & Bryan, 2010; Rhodes & Kates, 2015; Williams et al., 2008, 2012). Such findings are consistent with hedonic assumptions (i.e., pursuing pleasure and avoiding displeasure or pain), according to which positive (and regular) shifts in affective valence tend to increase the likelihood of future behavior, whereas negative shifts may have the opposite influence (Ekkekakis et al., 2011; Evmenenko & Teixeira, 2020; Rhodes & Kates, 2015).

Regarding exercise characteristics that may influence the affective response, exercise intensity stands out as the most relevant (Ekkekakis et al., 2011; Stevens et al., 2020). Current evidence suggests that participants exhibit distinct responses as intensity increases. Generally, during aerobic activities performed at intensities below the ventilatory threshold, most individuals either continue feeling good or feel even better than they did before exercise. At intensities proximal to the ventilatory threshold, inter-individual variability emerges, with some individuals feeling better but others feeling worse. Finally, at intensities that exceed the ventilatory threshold, there is a strongly homogeneous shift toward reduced pleasure (Ekkekakis et al., 2004; Ekkekakis et al., 2008). For resistance training, some evidence also indicates that increases in intensity (expressed as a percentage of one-repetition maximum, %1RM) are associated with increases in pleasure until the 70–80% 1RM interval, followed thereafter by an inverted association with pleasure (Andrade et al., 2022; Portugal et al., 2015), albeit in a magnitude and rhythm not yet known. Thus, targeting the intensity-pleasure/displeasure relation individually may be of value in the exercise domain when aiming to promote adherence (Teixeira et al., 2022), a topic that has been very scarcely explored via experimental studies.

1.2. Exercise prescription – a tripartite approach

A call to devise innovative exercise prescriptions aimed at pleasure promotion among diverse populations has been proposed by several authors (Ekkekakis et al., 2011; Ladwig et al., 2017; Teixeira et al., 2022). This entails an expansion of current exercise prescription guidelines, which are derived from dose-response relationships that only take into account effectiveness (e.g., fitness gains) and safety (e.g., reduced risk of injury). This bipartite or biomedical approach (e.g., adopted from the rationale followed in identifying the standard dose of pharmaceutical prescriptions) tends to overlook the fact that exercise is not a pill but a behavior, in which self-determined human beings must choose to partake voluntarily, frequently, and for the rest of their lives. Therefore, exercise prescriptions must take into account lessons from the behavioral sciences to encourage the long-term maintenance of this behavior (Kwasnicka et al., 2016), including, for example, the important issue of individual preferences. Although current exercise-prescription guidelines acknowledge the importance of taking individual choices into account, how to tailor training variables (such as exercise intensity) to individual characteristics is not specified in the guidelines and remains poorly explored.

The call for a tripartite exercise prescription that also incorporates the promotion of pleasure as a central consideration reflects the mounting evidence linking pleasurable experiences from exercise with adherence (Ekkekakis et al., 2011; Ladwig et al., 2017; Teixeira et al., 2022). Thus, besides offering programs that are effective and safe, contemplating how to promote pleasurable experiences from exercise is paramount. In this sense, some studies have shown that it is possible for participants to regulate exercise characteristics, and particularly intensity, when aiming to develop better affective responses, both in the laboratory and real-world settings (e.g., gymnasiums) (Costa et al., 2015; Elsangedy et al., 2018; Hamlyn-Williams et al., 2015; Parfitt et al., 2006; Rose & Parfitt, 2008), while improving physical health and exercise frequency (Parfitt et al., 2012; Williams et al., 2016). Indeed, in 2011, as stated on an ACSM position stand, affect regulation was recognized as an important consideration with the potential to facilitate adherence, although not specified as a primary method of exercise-intensity prescription. Nevertheless, the assessment of core affective valence assessment (e.g., through the Feeling Scale) was recommended as a method for the self-monitoring and self-regulation of exercise intensity besides heart rate and ratings of perceived exertion (Garber et al., 2011a).

The latest edition of the ACSM (2021) exercise-prescription guidelines reflect increased recognition of the potential value of affect-related behavioral strategies and theories, but still contain no specific instructions on how to best assess, promote, or use affect to regulate exercise intensity. Therefore, considerable work remains in this area. The void in present knowledge can be illustrated, for example, in the ACSM principles for exercise prescription (Frequency, Intensity, Time, and Type; FITT). Despite general recognition of the value of affect regulation for exercise promotion and maintenance, the FITT approach entails no explicit recommendation for any pre-exercise evaluation or in-session assessment of affect. Moreover, the FITT approach does not address how to adjust exercise prescriptions to improve the pleasure that participants derive from their exercise experience. In sum, general recognition of the value of affect regulation in exercise motivation is clear, but the ways of operationalizing it are lacking for proper advancement on this topic.

1.3. Objectives and hypothesis

Given the need to develop exercise prescriptions that can contribute to the improvement of adherence and the maintenance of regular physical activity, a Randomized Controlled Trial (RCT) targeting individual responses to exercise-associated pleasure in a health club setting will be conducted. The experimental protocol will compare two

individualized evaluation, prescription, and supervision methods focused on intensity, with the aim to determine the impact of promoting pleasure on behavioral, affective, and motivational outcomes.

The primary outcome of interest will be the gymnasium attendance during an eight-week follow-up period after an initial three-session intervention. Accordingly, the primary hypothesis that will be tested is that the experimental group will exhibit higher attendance during the follow-up period compared to the control group. In addition, the following secondary outcomes will be examined, hypothesizing that the results will favor the experimental group: (a) affective responses during the sessions, (b) post-exercise enjoyment, (c) intensity preference/tolerance agreement, (d) intention to continue exercising, (e) exercise-related habit, (f) core affective exercise experiences, (g) perceived choice, (h) perceived competence, (i) anticipated affective response, (j) remembered affect, (k) intrinsic motivation, and (l) subjective vitality. Finally, we also hypothesize that the slope of affect during the exercise session, the end-session affect, and the positive and negative affective peaks (the most pleasant and most unpleasant affect ratings during the session) will predict behavioral intention, anticipated affective responses, and remembered affect.

1.4. Trial design

The planned study will be a pragmatic randomized, single-blinded, controlled superiority trial with two parallel groups and an allocation ratio of 1:1. The study will be conducted between July and December of 2022. The individualized intervention will be implemented over two consecutive weeks. Forty-six individuals of both sexes will be enrolled and randomized in one of two arms. The control group (FITT) will receive a pre-exercise evaluation and three individualized training sessions based on the American College of Sports Medicine (ACSM) exercise-prescription guidelines and Frequency-Intensity-Time-Type (FITT) principles (ACSM, 2021). This is the approach commonly used to prescribe exercise for apparently healthy individuals that start to exercise in health clubs. As for the experimental group (FITT + AFFECT), attention will be given to exercise intensity assessment and manipulation, with the goal of optimizing the affective experience of exercise. The pre-exercise evaluation, the number of individualized sessions, and the methodological approach will be the same as that applied to the FITT group. However, additionally, individual preference and tolerance regarding exercise intensity will be assessed in the pre-exercise evaluation. These data will be used to select and advise on the intensity of the intervention exercise sessions. Moreover, individuals will be encouraged to continuously self-regulate their intensity to optimize their affective experience, aided by repeated assessments of their affective responses.

The trial has been registered on ClinicalTrials.gov with ID number NCT05416593. The SPIRIT recommendations (Chan et al., 2013) were followed for the development of the present protocol.

2. Method

2.1. Study setting

The intervention will take place at two health clubs in the Lisbon district of Portugal. One is a medium-sized mid-market club situated on the outskirts of Lisbon, and the other is a medium-sized club in the premium segment of the market located in the center of the city.

2.2. Eligibility criteria

To be eligible to participate in the trial, exercisers will need to (a) be between 18 and 45 years old, (b) be apparently healthy and without any contraindications to exercise, (c) have a body mass index (BMI) between 18.5 and 29.9 kg/m^2; (d) have previous experience in a gym or health club, and (e) have abstained from any type of regular strength, cardiovascular, or flexibility exercise program for at least 6 months (i.e., fewer than five workouts per month but no more than three months without any attendance) in a health club or any kind of structured sports activity. Exclusion criteria will disqualify individuals both prior to and during the intervention. Initial screening will disqualify prospective participants if they exhibit (a) 'high risk' stratification for cardiovascular disease (ACSM, 2021) and (b) high blood pressure ($\geq$130/80 mmHg). In addition, exclusion criteria during the intervention will include (c) acquired injury and (d) the inability to perform the exercise sessions (48h–96h apart) over two consecutive weeks.

2.3. Interventions

2.3.1. Pre-exercise evaluation and exercise program

The control group (FITT) will complete a pre-exercise evaluation that will assess general sociodemographic characteristics, psychological variables, body weight, blood pressure, and resting heart rate. The PAR-Q+ (Bredin et al., 2013) and cardiovascular disease risk assessment (ACSM, 2021) will also be part of this evaluation. The exercise program will follow the ACSM-FITT guidelines for apparently healthy adults beginning an exercise regimen (ACSM, 2021). FITT principles are presented in Table 1. The exercise sessions will occur on two days per week, with at least 48 h intervals and no more than 96 h apart. The sessions will be composed in the initial stage of 15 min of progressive aerobic activity on a treadmill (10 min at moderate intensity + 5 min at vigorous intensity); the fundamental stage will incorporate five resistance exercises (chest press, leg press, lat pulldown, leg curl, shoulder press) with 2 sets of 8–12 repetitions, 1 min rest between sets, and 2:0:2 cadence, followed by 15 min of decreasing aerobic-activity intensity on a stationary bicycle (5 min at vigorous intensity + 10 min at moderate intensity); the final stage will consist of four static passive stretching exercises (quadriceps, hamstrings, gluteus, triceps), performed with 2 sets of 20 s each. In this group, the intensity will be defined in accordance with the ACSM FITT principles (Table 1).

Table 1
Summary of experimental (FITT-AFFECT) and control (FITT) treatments.

	Control		
Frequency	Cardiovascular 2x/week	Resistance 2x/week	Flexibility 2x/week
Intensity	Moderate (45%–55% HRR) Vigorous (65%–75% HRR)	60%–70% RM;	Light-to-moderate
Time	30 min/session (20 min moderate; 10 min vigorous)	2 sets; 8–12 reps; 1 min rest	2x20 s
Type	Continuous	Isotonic; multijoint; large muscle groups	Static passive
	Experimental		
Frequency	Cardiovascular 2x/week	Resistance 2x/week	Flexibility 2x/week
Intensity	(1) Individualized according to preference/tolerance; (2) self-selected; (3) adjusted through FS/FAS		
Time	30 min; (1) Individualized according to preference/tolerance; (2) self-selected; (3) adjusted through FS/FAS	2 sets; 8–12 reps; 1 min rest	2x20 s
Type	Continuous	Isotonic; multijoint; large muscle groups	Static passive

Note. HRR = Heart Rate Reserve; RM = Repetition Maximum

The experimental group (FITT + AFFECT) will undergo the same pre-exercise evaluation and will be given the same exercise schedule as the FITT group. However, in the FITT + AFFECT group, the intensity will be individualized according to: (1) self-reported preference for and tolerance of exercise intensity (using a psychometric instrument that will be added to the pre-exercise assessment battery); (2) guided self-regulation of intensity with the goal of optimizing the affective experience of exercise; and (3) assessments of the core affective dimensions of valence and activation, using the Feeling Scale (FS) and Felt Arousal Scale (FAS), respectively. Specifically, intensity-preference scores will be used to set the initial intensity for cardiovascular, resistance, and flexibility exercises according to the stratification proposed by Bastos et al. (2022), namely light intensity: 5–9; light-to moderate-intensity: 10–14; moderate-to vigorous-intensity: 15–19; vigorous-intensity: 20–25. The definitions of these ranges of intensity will be aligned with the ACSM (2014, 2021) recommendations. For aerobic exercise, "light" intensity will be defined as < 60% of heart rate reserve (HRR), "light--to-moderate" as 60–69% HRR, "moderate-to-vigorous" as 70–85% HRR, and vigorous as > 85% HRR. For resistance exercise, "light" intensity will be defined as < 40% of 1-repetition maximum (1RM), "light--to-moderate" as 40–56% 1RM, "moderate-to-vigorous" as 60–79% 1RM, and "vigorous" > 80% 1RM. For stretching exercises, "light" intensity will be defined as up to the beginning of discomfort, "light-to-moderate" as inducing slight discomfort, "moderate-to-vigorous" as reaching the point of discomfort, and "vigorous" as inducing considerable discomfort and/or pain. Throughout the sessions, recommendations for self-regulating intensity will be repeated by the exercise professionals, encouraging participants to seek and maintain the intensity that they find most pleasurable for each exercise. Intensity-tolerance scores will also be used to advise individuals on the range of intensity that could be selected without immediate negative consequences for the quality of the affective experience. To facilitate the self-regulation of intensity, affective valence and activation will be measured repeatedly during the exercise sessions and used as the basis for intensity adjustments.

2.3.2. *Exercise session assessment time-points*

Psychometric measures will be administered before, during, and after the workouts in both groups (Figure 1). Pre-exercise assessments will take place 5–10 min before each session and will inquire about the anticipated affective response to the upcoming sessions and the remembered affective response to the previous session. During each session, heart rate (HR) will be measured continuously; the FS/FAS will be administered 15 times: at 5-min intervals during the initial aerobic activity (total of three-time points); immediately after the last set of each resistance exercise (total of five-time points) and at 5-min intervals during the aerobic activity during the main phase (total of three-time points); and in the last 10 s of the second set of each stretch in the final training stage (total of four-time points). Post-exercise assessments will take place at two-time points. The first assessment will be made 5 min after exercise and will evaluate the enjoyment of the preceding

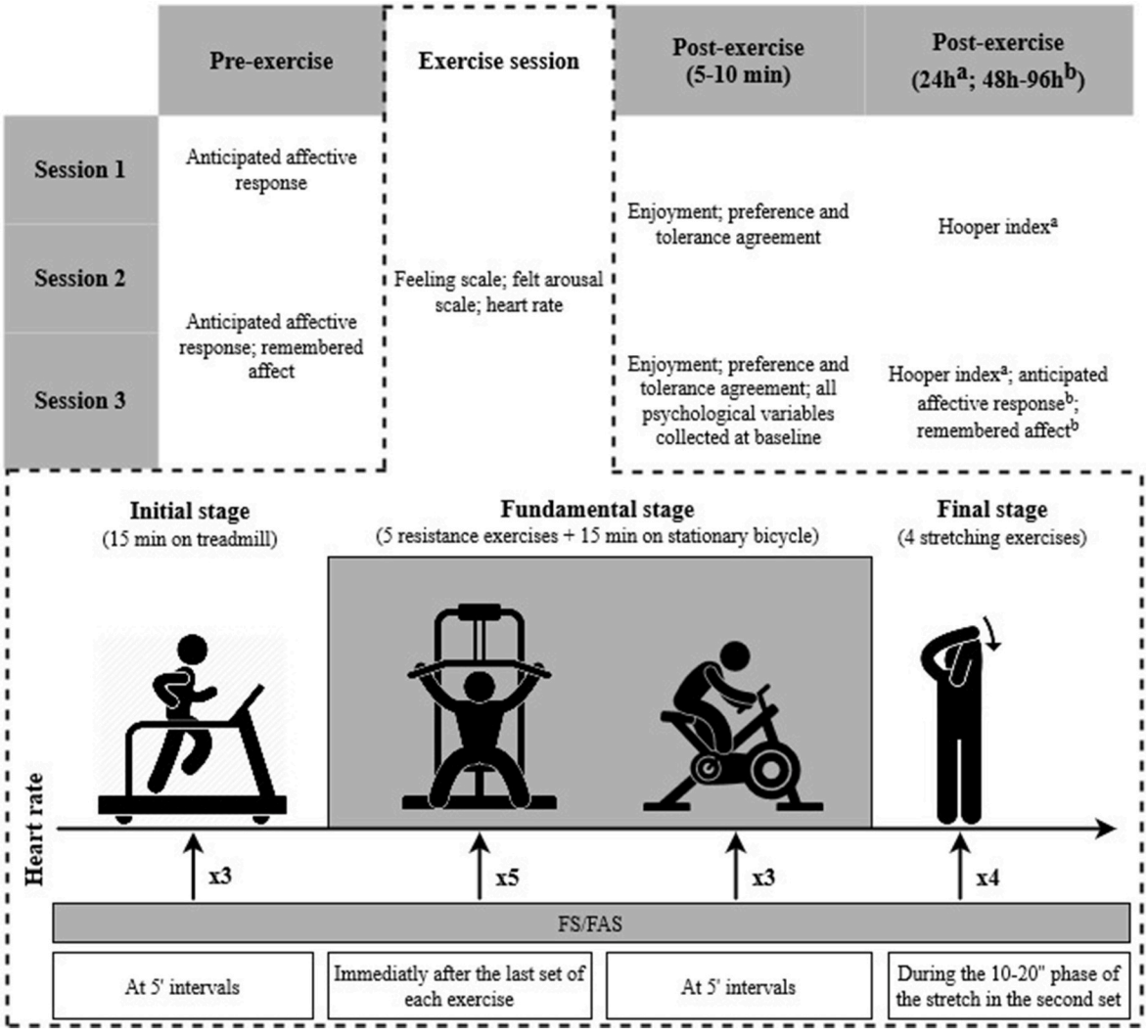


Figure 1. Timing protocol of assessment during the exercise sessions.

session and the level of agreement between the individual level of intensity-preference and intensity-tolerance with the intensity of the activity just completed. After the last session, an extended assessment will be conducted (including additional variables), as depicted in Figure 1. The second post-exercise assessment will take place 24 h after the end of the session, to determine the Hooper Index (see the section on Measures below for a detailed description). In addition, the anticipated affective response and remembered affect will be assessed 48 h–96 h after the last session, to mimic the typical interval between exercise sessions.

Any qualitative information informally and spontaneously provided by the participants regarding their mood, stress, sleep, or any perturbation of their daily routine that can influence their exercise will be registered. Reported injury or illness throughout the study will be analyzed and may result in the termination of individual participation if the exerciser's health or the protocol implementation is compromised. Changes reported in the baseline assessment regarding inclusion criteria (e.g., blood pressure) will be analyzed and may result in the interruption of the individual trial. Participant request to halt an exercise or the trial will be immediately accepted.

2.4. Outcomes

As noted earlier, the primary outcome will be exercise attendance during the eight weeks following the intervention. Comparisons will be made across time by group and between groups on a weekly basis. Secondary outcomes will include affect-related variables (i.e., affective responses, anticipated affective response, remembered affective response, enjoyment) and motivational variables (i.e., intrinsic motivation, core affective exercise experiences, intention to continue exercising, exercise habit, perceived choice, perceived competence, subjective vitality). Other variables (e.g., heart rate and perceptions of effort) will be used for control purposes. Given the nature of the interventions and outcome measures, no harm to the participants is expected.

2.5. Sample size

Power calculations for the determination of sample size were performed using G*Power (v.3.1.9.7; Faul, Erdfelder, Buchner, & Lang, 2009). For the statistical analysis to be performed on the primary outcome, a split-plot ANOVA (2 groups x 8 time-points), anticipating a "small" to "medium" effect size for the interaction ($f = 0.20$), with an $\alpha = 0.05$, statistical power ($1 - \beta$) of 0.90, a correlation between the repeated measurements of $r = 0.7$, and a violation of sphericity ($\varepsilon = 0.7$), will require a total sample size of 24 individuals (12 per group). For the analysis of the secondary outcomes, and when considering the most restrictive analysis to be performed (i.e., split-plot ANOVA (2 groups x 2 time-points)) (see Statistical Methods section), the a priori sample size calculations indicate the need for a total of 42 individuals (21 per group). Therefore, with an oversampling of ~10% to account for dropout, a total of 46 participants (23 per group) will be recruited, to provide sufficient statistical power.

2.6. Recruitment and allocation

A rolling recruitment will be used. A call for participants will be issued through adverts on online social media, around a university campus, and especially in the health clubs participating in the study. Initial screening will be made by the researchers to ensure that interested individuals meet the eligibility criteria. Informed consent, outlining the research line general purposes, expectations, and risks, will be obtained before enrollment into the study. All individuals that fulfill the eligibility criteria (until reaching the goal of 46 participants) will be offered a pre-exercise evaluation and three individualized exercise sessions with accredited exercise physiologists. A brief report summarizing the study results will be made available to the exercisers at the end of the study if requested. Randomization to the control (FITT) or experimental (FITT + AFFECT) groups will be through random assignment using computer software after the initial screening, conducted by a researcher not involved in data collection. The participants will be blinded to the intervention they receive (i.e., will not be told that they were assigned to the "experimental" or the "control" group), to minimize expectations and the Hawthorne effect. The study flowchart (Figure 2) depicts the process for screening and eligibility, randomization, pre-exercise assessment, and exercise protocols.

2.7. Data collection methods

2.7.1. During-exercise affective responses

The core affective responses to exercise will be assessed with the Feeling Scale (FS; Hardy & Rejeski, 1989) and Felt Arousal Scale (FAS; Svebak & Murgatroyd, 1985) Portuguese versions (Brito et al., 2022), which are theoretically proposed to evaluate the core affective dimensions of valence and activation, respectively. The FS is an 11-point scale ranging from −5 (*"Very bad"*) to +5 (*"Very good"*), which assesses the affective valence; the FAS is a 6-point scale ranging from 1 (*"Low arousal"*) to 6 (*"High arousal"*) that assesses perceived activation.

2.7.2. Reflective affect processing

2.7.2.1. Core affective exercise experiences and physical activity enjoyment. Core affective experiences associated with past exercise participation will be measured using four items from the Affective Exercise Experiences (AFFEXX) questionnaire (Ekkekakis et al., 2021) that evaluate the pleasure-displeasure factor of the core affective experience. Responses are given on a 7-point Likert scale with antithetically worded statements (*"exercise makes me feel worse – exercise makes me feel better"*). The AFFEXX has been translated to Portuguese using Brislin's (1970, 1980) recommendations.

Additionally, perceptions of the degree of exercise enjoyment will be assessed using an instrument composed of eight items (e.g., *"It is fun"*) answered on a 7-point Likert scale ranging from 1 (*"Totally disagree"*) to 7 (*"Totally agree"*) following similar studies in this context the (e.g., Rodrigues et al., 2020; Teques et al., 2017).

2.7.2.2. Anticipated affective response. The Empirical Valence Scale (EVS; Lishner et al., 2008), adapted as suggested in a related study (Zenko et al., 2016), will be used to measure anticipated/forecasted affective response to the exercise sessions. This instrument uses 15 empirically spaced verbal anchors ranging from −100 (*"Most unpleasant imaginable"*) to +100 (*"Most pleasant imaginable"*). Participants will respond to the question *"Considering the workout you are about to begin* [previously explained/remembered]*, how do you think it will make you feel?"*. The EVS will be used to reduce common-method bias with the FS (to be used for the assessment of a different construct, namely momentary affective valence).

2.7.2.3. Remembered affect. A Visual Analog Scale (VAS) will be used to assess remembered exercise pleasure/displeasure, as suggested in a similar study (Zenko et al., 2016). As was also the case for the EVS, the VAS will be used in order to reduce common-method bias with the FS. This scale ranges from +100 (*"very pleasant"*) to −100 (*"very unpleasant"*). Prior to the exercise sessions, participants will be asked to rate *"How did your last workout make you feel?"*.

2.7.3. Intensity preference, tolerance, and agreement

The Preference for and Tolerance of Exercise Intensity Questionnaire (PRETIE-Q; Ekkekakis et al., 2005; Portuguese version: Teixeira et al., 2021) will be used to assess individual differences in intensity-preference, intensity-tolerance, and the agreement between

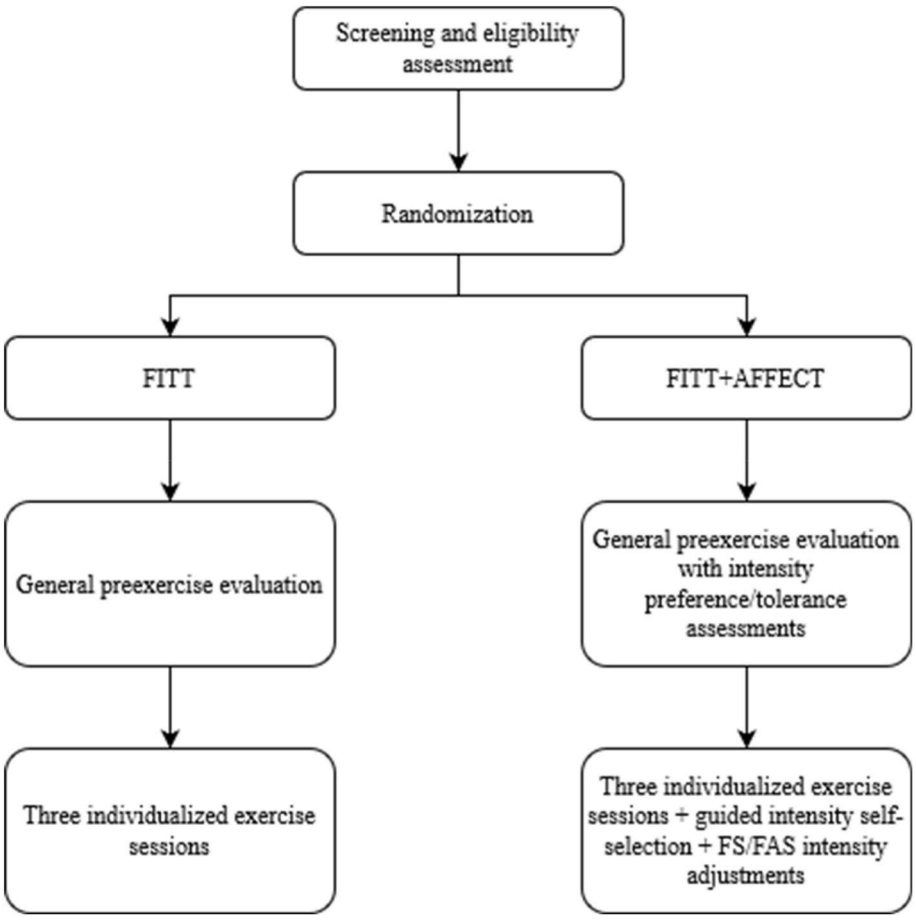


Figure 2. Study flow-chart.

self-reported preference and tolerance with the intensity experienced during the preceding exercise session. The instrument is composed of 10 items (five for each factor; preference: *"Low-intensity exercise is boring"*; tolerance: *"Feeling tired during exercise is my signal to slow down or stop"*) answered on a 5-point Likert scale ranging from 1 (*"Totally disagree"*) to 5 (*"Totally agree"*). The two additional questions proposed by this instrument version to assess agreement will also be used.

2.7.4. Intrinsic motivation, perceived choice, and perceived competence

Three subscales (intrinsic motivation, perceived choice, and perceived competence) from the Intrinsic Motivation Inventory (IMI; McAuley et al., 1989; Portuguese version: Fonseca & de Paula Brito, 2001) will be used. Minor adaptations to the wording will be made to ensure relevance to the context of exercise (e.g., vs. sport) following the example of adaptations in similar studies (e.g., Vazou-Ekkekakis & Ekkekakis, 2009; Zenko et al., 2016). For example, *"I am satisfied with my performance at this task"* (perceived competence) will be adapted to *"I am satisfied with my performance in these workouts";* and *"I believe I had some choice about doing this activity"* (perceived choice) will be adapted to *"I believe I had some choice during these workouts"*. One item will be added to the perceived choice subscale regarding the freedom to select exercise intensity, i.e., *"I felt I had the choice to switch the intensity of exercise as I wished"*, as suggested in a similar study (Vazou-Ekkekakis & Ekkekakis, 2009). Each of the three subscales has 7 items (8 in perceived choice with the addition of the extra item), each rated on a 5-point scale ranging from 1 *("Totally disagree")* to 5 *("Totally agree")*.

2.7.5. Subjective vitality

The Subjective Vitality Scale (SVS; Ryan & Frederick, 1997; Portuguese version: Moutão et al., 2013) will be used to assess subjective vitality, a manifestation of enthusiasm and inspiration usually experienced as a state of feeling alive and with energy. It consists of six items (e.g., *"I feel alive and vitalized"*), answered using a 7-point bipolar scale, ranging from 1 (*"Totally disagree"*) to 7 *("Totally agree"*).

2.7.6. Intention to continue exercising

The behavioral intention will be assessed using three items (e.g., *"I will continue to practice physical exercise in the next* 6 months *as I recently practiced* [with or without personal trainer] *or in a very similar way (same type, frequency, duration, and intensity"*) previously used in the exercise context for the Portuguese population (Rodrigues et al., 2020) and grounded in recommendations by Ajzen (2006). Minor adjustments related to the use of a personal training service were added (square brackets), to avoid bias in the responses due to personal financial limitations and to frame the personal choice as related to the activity and not the supervision by an exercise professional. Responses are entered using a 7-point bipolar scale ranging from 1 (*"Absolutely not"*) to 7 (*"Absolutely yes"*).

2.7.7. Exercise habit

The Self-Report Behavioral Automaticity Index (SRBAI; Gardner et al., 2011b; Portuguese version: Rodrigues et al., 2021a) will be used to assess behavioral habit development. The instrument consists of four items (e.g., *"I do without thinking"*) answered on a 7-point Likert scale, ranging from 1 (*"Totally disagree"*) to 7 (*"Totally agree"*).

Training control variables.

2.7.8. Heart rate

As a manipulation check, heart rate will be monitored throughout the session using a Polar H10 (Polar Electro Inc., Kempele, Finland)

chest sensor paired to the Elite HRV app (Version: 5.5.4; Gloucester, Massachusetts), validated in field testing (Perrotta et al., 2017).

2.7.9. Blood pressure

Blood pressure will be tested in the pre-exercise evaluation session using an OMRON M3 Intellisense HEM7051-E (OMRON Healthcare Europe, B.V., Hoofddorp, The Netherlands) following ACSM (2021) recommendations, and values of 130/80 mmHg used as a cut-off for study enrollment.

2.7.10. Weight

Body weight will be measured using a TANITA SC-330 (TANITA Europe B.V., Amsterdam, The Netherlands) following ACSM (2021) recommendations. Height data will be obtained from identification cards and combined with weight data to determine BMI.

2.7.11. Sleep, stress, fatigue, and muscle soreness (Hooper Index)

The Hooper questionnaire (Hooper & Mackinnon, 1995) will be used to control and compare 24-h post-exercise subjective perceptions of the impact of each exercise session on four dimensions: sleep, stress, fatigue, and muscle soreness (Delayed-Onset Muscle Soreness; DOMS). For each factor, exercisers will rate their perceptions on a scale ranging from 1 to 7. For sleep, verbal descriptors will present the terms *"Very, very good - Very, very bad"*; for stress, fatigue, and DOMS, the anchors range from *"Very, very low" to "Very, very high"*. The sum of all categories is represented as the Hooper index.

No plans to promote participant retention and complete follow-up will be developed. Outcome data from participants who discontinue their participation will not be considered for future statistical analysis.

2.8. Data management

Data will be collected by two researchers implementing the interventions, and data entry will be carried out by two other researchers. The dataset will be stored in a private server in accordance with the General Data Protection Regulation (GDPR), and the data will be anonymized. Regular checks (weekly; throughout intervention and data collection) will be made to ensure data quality and integrity.

2.9. Statistical methods

Preliminary tests will be performed to test for possible violations of basic statistical assumptions. All tests will be performed with SPSS v.27.0 (IBM, Co.)

2.10. Primary outcome

The main hypothesis refers to differences in exercise attendance between groups and across eight weeks following the intervention, measured weekly through electronic turnstile access. Therefore, a split-plot ANOVA (2 groups x 8 time-points) will be conducted.

2.11. Secondary outcomes

For the comparison between affective responses per session, a split-plot ANOVA (2 groups x 6 time-points) will be performed separately for FS and FAS. Of the 15 assessment time points included to the facilitation of the affect-based self-regulation of intensity, six will be used for analysis, based on recent recommendations in a similar exercise program (Bastos et al., 2022). The time points used in the analysis will be the 10th and 15th min of the aerobic activity during the initial stage (thus capturing the end of the moderate and vigorous intensities in the control group; while the activity is ongoing), the final measure of the lateral pulldown (approximately at the middle of the five resistance exercises), the 5th and 15th min of the aerobic activity (at the end of the vigorous and moderate intensities in the control group; while the activity is ongoing) during the fundamental stage, and the last measurement taken during the stretching activities in the training final stage. A 2 (groups) x 3 (sessions) split-plot ANOVA will be performed to compare end-session enjoyment between groups and across the sessions. A 2 (groups) x 2 (time points) split-plot ANOVA will be performed for core affective exercise experiences. A split-plot ANOVA will be performed for remembered affect (2 groups x 3 sessions) and for anticipated affective responses (2 groups x 4 time-points). Finally, a series of regression analyses will be conducted to determine if (a) the slope of affect during the sessions, (b) the reported end-session affect, and (c) the positive and negative affective peaks, predict the anticipated affective responses and remembered affect.

2.12. Other outcomes

The intensity preference/tolerance agreement in each group will be examined with a multiple logistic regression analysis. A 2 (groups) x 2 (time points) split-plot ANOVA will be used for each motivational construct (intention to continue exercising, exercise-related habit, perceived choice and competence, intrinsic motivation, and subjective vitality). Regression methods will be used to test whether data from the last session predict behavioral intention. Any additional analyses may be reported as descriptive or exploratory.

3. Ethics and dissemination

The present trial was registered with ID number NCT05416593. Approval was obtained by the institutional ethics committee before the study's commencement. The study will be conducted in accordance with international conventions and declarations related to ethical research practices (World Medical Association, 2013). Participants will be asked to give consent for study participation and data collection in the recruitment stage and will be able to interrupt or discontinue their participation at any time without repercussions. Study enrollment and participation will halt if adverse events are detected, in accordance with international guidelines (Medical Research Council, 2017). If protocol amendments are deemed necessary, additional approval from the institutional ethics committee will be sought and the participants will be informed of the modifications and their possible impact. The trial results will be published in peer-review scientific journals. Preliminary, intermediate, or secondary results may be presented at national and international scientific events. Baseline data will be analyzed only after the end of the intervention period. Coded and anonymized final datasets will be in the care of authors DT and ALP. Authorship will be determined in accordance with guidelines set forth by the International Committee of Medical Journal Editors. An anonymized global dataset may be made available upon reasonable request under the supervision of one of the protocol authors. The authors declare no conflict of interest.

4. Potential outcomes and impact

The need to improve population-level physical activity rates has prompted researchers to consider the potential motivational implications of the affective experiences people derive from their physical activity and exercise participation, such as pleasure versus displeasure. However, the current evidence base is still limited to cross-sectional and longitudinal observational research (Ekkekakis et al., 2019). Therefore, there is a pressing need for experimental designs that will allow the testing of theoretical predictions and, ultimately, the advancement of current exercise prescription guidelines and applied practice in real-world settings.

The present RCT protocol addresses part of this gap, by testing the effects of manipulating exercise intensity (i.e., using standard guideline-based prescription versus encouraging affect-oriented self-regulation) on short-term exercise frequency, as well as on a wide array of affect-related variables (i.e., affective response, remembered affect,

anticipated affect, affective exercise experiences, enjoyment) that have been found to predict motivational variables (e.g., intention, habit) and exercise behavior (Rhodes & Kates, 2015; Stevens et al., 2020; Strobach et al., 2020; Teixeira et al., 2022; Weyland et al., 2020).

An additional contribution pertains to highlighting the importance of individuality in exercise prescriptions. Robust evidence demonstrates that the ways people experience pleasurable responses are inherently subjective and, therefore, individually targeting positive affective experiences and enjoyment could be a core mechanism for promoting exercise participation and adherence (Chen et al., 2020; Ekkekakis et al., 2011; Slawinska & Davis, 2020; Teixeira et al., 2022; Weyland et al., 2020). It is expected that by taking into account individual differences in intensity preference and tolerance, the exercise sessions would approximate what would be individually perceived as pleasurable, as demonstrated in related studies (Bastos et al., 2022; Box & Petruzzello, 2020). As an additional method of harnessing individuality, the guided self-regulation of intensity will be promoted as a way of enhancing the affective experience of exercise. First, this approach is based on the assumption that not all exercisers (particularly those with low attendance; at risk of dropout; those with low experience) are aware of their physiological limits or options for adjusting their intensity in a variety of exercises, activities, or exercise modalities (Marques et al., 2022; Rose & Parfitt, 2012; Teixeira et al., 2021). Thus, intensity preference, and especially intensity tolerance, will be considered when advising participants to pursue intensity adjustments, bearing in mind that low tolerance scores would indicate the need for minor intensity adjustments, whereas higher tolerance scores would suggest a wider range of possibilities without necessarily incurring immediate negative consequences for the affective experience. Secondly, aiming for pleasure-oriented self-regulation of intensity is different from *just* the self-regulation of intensity (Zenko et al., 2017). Both are predicated on the idea that choice matters in promoting pleasure and enjoyment. This is based on several studies showing that imposed exercise intensities tend to decrease interest and enjoyment due to a perceived loss or restriction of autonomy, with negative consequences for perceived competence and intrinsic motivation (Parfitt et al., 2006; Rose & Parfitt, 2012; Vazou-Ekkekakis & Ekkekakis, 2009). It is expected that most individuals, when given the opportunity to self-regulate their intensity, will choose an intensity that is perceived as pleasurable (Ekkekakis, 2009; Zenko et al., 2017). However, the distinct contribution of providing participants with perceived choice and encouraging them to regulate their intensity with the express goal of optimizing their affective experience remains unexplored. Although conceivably more effective when combined, uncovering their individual role and their relative importance may be important for tailoring exercise interventions to individuals. All in all, these suggestions (i.e., providing exercise options that are enjoyable; facilitating the self-regulation of behavior; providing support) align with contemporary approaches and recommendations for health behavior change and maintenance (Dukes et al., 2021; Kwasnicka et al., 2016; Ntoumanis et al., 2018; Rodrigues et al., 2019), and may present valuable opportunities for the improvement of exercise prescription guidelines and professional practice norms.

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Authors' contributions

DT and PE conceived the study. AA, VB, and DT are responsible for the study design and implementation. DT and ALP are responsible for the primary statistical analysis and study supervision. All authors contributed to the refinement of the study protocol and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.psychsport.2023.102424.

References

Ajzen, I. (2006). *Constructing a TPB questionnaire: Conceptual and methodological considerations*. University of Massachusetts.

American College of Sports Medicine. (2014). *ACSM's guidelines for exercise testing and prescription*. Lippincott Williams & Wilkins.

American College of Sports Medicine. (2021). *ACSM's guidelines for exercise testing and prescription*. Lippincott Williams & Wilkins.

Andrade, A., Ekkekakis, P., Evmenenko, A., Monteiro, D., Rodrigues, F., Cid, L., & Teixeira, D. S. (2022). Affective responses to resistance exercise: Toward consensus on the timing of assessments. *Psychology of Sport and Exercise, 62*, Article 102223. https://doi.org/10.1016/j.psychsport.2022.102223

Bastos, V., Andrade, A. J., Rodrigues, F., Monteiro, D., Cid, L., & Teixeira, D. S. (2022). Set to fail: Affective dynamics in a resistance training program designed to reach muscle concentric failure. *Scandinavian Journal of Medicine & Science in Sports, 32* (12), 1710–1723. https://doi.org/10.1111/sms.14222

Box, A., & Petruzzello, S. (2020). Why do they do it? Differences in high-intensity exercise-affect between those with higher and lower intensity preference and tolerance. *Psychology of Sport and Exercise, 47*, Article 101521. https://doi.org/10.1016/j.psych sport.2019.04.011

Brand, R., & Ekkekakis, P. (2018). Affective–reflective theory of physical inactivity and exercise. *German Journal of Exercise and Sport Research, 48*(1), 48–58. https://doi.org/10.1007/s12662-017-0477-9

Bredin, S. S., Gledhill, N., Jamnik, V. K., & Warburton, D. E. (2013). PAR-Q+ and ePARmed-X+: New risk stratification and physical activity clearance strategy for physicians and patients alike. *Canadian Family Physician, 59*(3), 273–277.

Brislin, R. W. (1970). Back-translation for cross-cultural research. *Journal of Cross-Cultural Psychology, 1*(3), 185–216. https://doi.org/10.1177/135910457000100301

Brislin, R. W. (1980). Translation and content analysis of oral and written material. In H. C. Triandis, & R. W. Brislin (Eds.), *Handbook of cross-cultural psychology* (pp. 389–444). Allyn and Bacon.

Brito, H., Teixeira, D. S., & Araujo, D. (2022). Construct validity of the feeling scale and felt arousal scale in Portuguese Recreational Exercisers. *Cuadernos de Psicología del Deporte, 22*, 103–113. https://doi.org/10.6018/cpd.514061

Buckworth, J., Dishman, R., O'Connor, P., & Tomporowski, P. (2013). Exercise psychology. In *Human kinetics* (2nd ed.).

Chan, A. W., Tetzlaff, J. M., Altman, D. G., Laupacis, A., Gøtzsche, P. C., Krleža, J., & Moher, D. (2013). SPIRIT 2013 statement: Defining standard protocol items for clinical trials. *Annals of Internal Medicine, 158*(3), 200–207. https://doi.org/10.7326/0003-4819-158-3-201302050-00583

Chen, C., Finne, E., Kopp, A., & Jekauc, D. (2020). Can positive affective variables mediate intervention effects on physical activity? A systematic review and meta-analysis. *Frontiers in Psychology, 11*, Article 587757. http://doi:10.3389/fpsyg.2020.587757.

Cheval, B., & Boisgontier, M. P. (2021). The theory of effort minimization in physical activity. *Exercise and Sport Sciences Reviews, 49*(3), 168–178. https://doi.org/10.1249/JES.0000000000000252

Costa, E. C., de Sá, J. C., Costa, I. B., Meireles, R., Lemos, T. M., Elsangedy, H. M., Krinski, K., & Azevedo, G. D. (2015). Affect-regulated exercise: An alternative approach for lifestyle modification in overweight/obese women with polycystic ovary syndrome. *Gynecological Endocrinology, 31*(12), 971–975. https://doi.org/10.3109/09513590.2015.1092132

Dukes, D., Abrams, K., Adolphs, R., Ahmed, M. E., Beatty, A., Berridge, K. C., & Clay, Z. (2021). The rise of affectivism. *Nature Human Behaviour, 5*(7), 816–820. https://doi.org/10.1038/s41562-021-01130-8

Ekkekakis, P. (2009). Let them roam free? Physiological and psychological evidence for the potential of self-selected exercise intensity in public health. *Sports medicine (Auckland, N.Z.), 39*(10), 857–888. https://doi.org/10.2165/11315210-000000000-00000.

Ekkekakis, P. (2013). *The measurement of affect, mood, and emotion: A guide for health-behavioral research*. Cambridge University Press. https://doi.org/10.1017/CBO9780511820724

Ekkekakis, P., Hall, E. E., & Petruzzello, S. J. (2004). Practical markers of the transition from aerobic to anaerobic metabolism during exercise: Rationale and a case for affect-based exercise prescription. *Preventive Medicine, 38*(2), 149–159. http://doi:10.1016/j.ypmed.2003.09.038.

Ekkekakis, P., Hall, E., & Petruzzello, S. (2005). Some like it vigorous: Measuring individual differences in the preference for and tolerance of exercise intensity. *Journal of Sport & Exercise Psychology, 27*. https://doi.org/10.1123/jsep.27.3.350

Ekkekakis, P., Hall, E. E., & Petruzzello, S. J. (2008). The relationship between exercise intensity and affective responses demystified: To crack the 40-year-old nut, replace the 40-year-old nutcracker. *Annals of Behavioral Medicine, 35*(2), 136–149. https://doi.org/10.1007/s12160-008-9025-z

Ekkekakis, P., Hartman, M. E., & Ladwig, M. A. (2019). Conceptual foundations of exercise psychology: Facilitators, inhibitors, and a roadmap towards establishing societal relevance. In M. H. Anshel (Ed.), *APA handbook of sport and exercise psychology, 2* pp. 27–56). Washington, DC: American Psychological Association. https://doi.org/10.1037/0000124-002.

Ekkekakis, P., Parfitt, G., & Petruzzello, S. J. (2011). The pleasure and displeasure people feel when they exercise at different intensities. *Sports Medicine, 41*(8), 641–671. https://doi.org/10.2165/11590680-000000000-00000

Ekkekakis, P., & Petruzzello, S. J. (2000). Analysis of the affect measurement conundrum in exercise psychology: IV. A conceptual case for the affect circumplex. *Psychology of Sport and Exercise, 3*(1), 35–63. https://doi.org/10.1016/S1469-0292(01)00028-0

Ekkekakis, P., & Zenko, Z. (2016). Escape from cognitivism: Exercise as hedonic experience. In M. Raab, P. Wylleman, R. Seiler, A.-M. Elbe, & A. Hatzigeorgiadis (Eds.), *Sport and exercise psychology research: From theory to practice* (pp. 389–414). Elsevier Academic Press. https://doi.org/10.1016/B978-0-12-803634-1.00018-2.

Ekkekakis, P., Zenko, Z., & Vazou, S. (2021). Do you find exercise pleasant or unpleasant? The affective exercise experiences (AFFEXX) questionnaire. *Psychology of Sport and Exercise, 55*, Article 101930. https://doi.org/10.1016/j.psychsport.2021.101930

Elsangedy, H. M., Machado, D. G., Krinski, K., Nascimento, P. H., Oliveira, G. T., Santos, T. M., Hargreaves, E. A., & Parfitt, G. (2018). Let the pleasure guide your resistance training intensity. *Medicine & Science in Sports & Exercise, 50*(7), 1472–1479. https://doi.org/10.1249/MSS.0000000000001573

European Commission. (2022). *Special Eurobarometer 525 report – sport and physical activity*. European Commission.

Evmenenko, A., & Teixeira, D. S. (2020). The circumplex model of affect in physical activity contexts: A systematic review. *International Journal of Sport and Exercise Psychology*, 1–34. https://doi.org/10.1080/1612197X.2020.1854818

Faul, F., Erdfelder, E., Buchner, A., & Lang, A. G. (2009). Statistical power analyses using G*power 3.1: Tests for correlation and regression analyses. *Behavior Research Methods, 41*(4), 1149–1160. https://doi.org/10.3758/BRM.41.4.1149

Fonseca, A. M., & de Paula Brito, A. (2001). Propriedades psicométricas da versão portuguesa do Intrinsic Motivation Inventory (IMIp) em contextos de actividade física e desportiva. *Análise Psicológica, 19*(1), 59–76.

Garber, C. E., Blissmer, B., Deschenes, M. R., Franklin, B. A., Lamonte, M. J., Lee, I.-M., … Swain, D. P. (2011a). American College of sports medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: Guidance for prescribing exercise. *Medicine & Science in Sports & Exercise, 43*(7), 1334–1359. https://doi.org/10.1249/MSS.0b013e318213fefb

Gardner, B., Abraham, C., Lally, P., & Brujin, G. (2011b). Towards parsimony in habit measurement: Testing the convergent and predictive validity of an automaticity subscale of the Self-Report Habit Index. *International Journal of Behavioral Nutrition and Physical Activity, 9*(102), 1–12. https://doi.org/10.1186/1479-5868-9-102

Hamlyn-Williams, C. C., Tempest, G., Coombs, S., & Parfitt, G. (2015). Can previously sedentary females use the feeling scale to regulate exercise intensity in a gym environment? An observational study. *BMC Sports Science, Medicine & Rehabilitation, 7*(30). https://doi.org/10.1186/s13102-015-0023-8

Hardy, C. J., & Rejeski, W. J. (1989). Not what, but how one feels: The measurement of affect during exercise. *Journal of Sport & Exercise Psychology, 11*(3), 304–317. https://doi.org/10.1123/jsep.11

Hooper, S. L., & Mackinnon, L. T. (1995). Monitoring overtraining in athletes. *Sports Medicine, 20*(5), 321–327. http://doi:10.2165/00007256-199520050-00003.

International Health, Racquet & Sports Club Association. (2020). The 2020 IHRSA global report: The state of the health club industry. https://www.ihrsa.org/publications/the-2020-ihrsaglobal-report. (Accessed 12 January 2021).

Kwan, B. M., & Bryan, A. (2010). In-task and post-task affective response to exercise: Translating exercise intentions into behaviour. *British Journal of Health Psychology, 15*(Pt 1), 115–131. https://doi.org/10.1348/135910709X433267

Kwasnicka, D., Dombrowski, S. U., White, M., & Sniehotta, F. (2016). Theoretical explanations for maintenance of behaviour change: A systematic review of behaviour theories. *Health Psychology Review, 10*(3), 277–296. https://doi.org/10.1080/17437199.2016

Ladwig, M. A., Hartman, M. E., & Ekkekakis, P. (2017). Affect-based exercise prescription. *ACSM's Health & Fitness Journal, 21*(5), 10–15. https://doi.org/10.1249/FIT.0000000000000332

Lishner, D. A., Cooter, A. B., & Zald, D. H. (2008). Addressing measurement limitations in affective rating scales: Development of an empirical valence scale. *Cognition & Emotion, 22*(1), 180–192. https://doi.org/10.1080/02699930701319139

Marques, P., Andrade, A. J., Evmenenko, A., Monteiro, D., Faria, J., Rodrigues, F., Cid, L., & Teixeira, D. S. (2022). The preference for and tolerance of exercise intensity: An exploratory analysis of intensity discrepancy in health clubs settings. *Current Psychology*. https://doi.org/10.1007/s12144-022-03149-0

McAuley, E., Duncan, T., & Tammen, V. V. (1989). Psychometric properties of the intrinsic motivation inventory in a competitive sport setting: A confirmatory factor analysis. *Research Quarterly for Exercise & Sport, 60*(1), 48–58. https://doi.org/10.1080/02701367.1989.10607413

Medical Research Council. (2017). MRC guidelines for management of global health trials - involving Clinical or Public Health Interventions. Available: https://mrc.ukri.org/documents/pdf/guidelines-for-management-of-global-health-trials/. (Accessed 18 March 2020).

Moutão, J. M., Mendes Alves, S., & Cid, L. (2013). Translation and validation of the subjective vitality scale in a Portuguese sample of exercise participants. *Revista Latinoamericana de Psicología, 45*(2), 223–230.

Ntoumanis, N., Thøgersen-Ntoumani, C., Quested, E., & Chatzisarantis, N. (2018). Theoretical approaches to physical activity promotion. In O. Braddick (Ed.), *Oxford research encyclopaedia of psychology*. Oxford University Press.

Parfitt, G., Alrumh, A., & Rowlands, A. V. (2012). Affect-regulated exercise intensity: Does training at an intensity that feels 'good' improve physical health? *Journal of Science and Medicine in Sport, 15*(6), 548–553. https://doi.org/10.1016/j.jsams.2012.01.005

Parfitt, G., Rose, E. A., & Burgess, W. M. (2006). The psychological and physiological responses of sedentary individuals to prescribed and preferred intensity exercise. *British Journal of Health Psychology, 11*(Pt 1), 39–53. https://doi.org/10.1348/135910705X43606

Perrotta, A. S., Jeklin, A. T., Hives, B. A., Meanwell, L. E., & Warburton, D. (2017). Validity of the elite HRV smartphone application for examining heart rate variability in a field-based setting. *The Journal of Strength & Conditioning Research, 31*(8), 2296–2302. https://doi.org/10.1519/JSC.0000000000001841

Portugal, E. M., Lattari, E., Santos, T. M., & Deslandes, A. C. (2015). Affective responses to prescribed and self-selected strength training intensities. *Perceptual & Motor Skills, 121*(2), 465–481. https://doi.org/10.2466/29.PMS.121c17x3

Rhodes, R. E., & Kates, A. (2015). Can the affective response to exercise predict future motives and physical activity behavior? A systematic review of published evidence. *Annals of Behavioral Medicine, 49*(5), 715–731. https://doi.org/10.1007/s12160-015-9704-5

Rhodes, R. E., McEwan, D., & Rebar, A. L. (2019). Theories of physical activity behaviour change: A history and synthesis of approaches. *Psychology of Sport and Exercise, 42*, 100–109. https://doi.org/10.1016/j.psychsport.2018.11.010

Rodrigues, F., Cid, L., Forte, P., Teixeira, D., Travassos, B., & Monteiro, D. (2021a). A avaliação do hábito em praticantes de exercício físico: Testando a validade do self-report behavioral automaticity index. *Motricidade, 17*(2), 185–192. https://doi.org/10.6063/motricidade.21495

Rodrigues, F., Teixeira, D. S., Cid, L., & Monteiro, D. (2019). Promoting physical exercise participation: The role of interpersonal behaviors for practical implications. *Journal of Functional Morphology and Kinesiology, 4*(2), 40. https://doi.org/10.3390/jfmk4020040

Rodrigues, F., Teixeira, D., Neiva, H., Cid, L., & Monteiro, D. (2020). The bright and dark sides of motivation as predictors of enjoyment, intention, and exercise persistence. *Scandinavian Journal of Medicine & Science in Sports, 30*(4), 787–800. https://doi.org/10.1111/sms.13617

Rose, E. A., & Parfitt, G. (2008). Can the Feeling Scale be used to regulate exercise intensity? *Medicine & Science in Sports & Exercise, 40*(10), 1852–1860. https://doi.org/10.1249/MSS.0b013e31817a8aea

Rose, E. A., & Parfitt, G. (2012). Exercise experience influences affective and motivational outcomes of prescribed and self-selected intensity exercise. *Scandinavian Journal of Medicine & Science in Sports, 22*(2), 265–277. https://doi.org/10.1111/j.1600-0838.2010.01161.x

Ryan, R. M., & Frederick, C. (1997). On energy, personality, and health: Subjective vitality as a dynamic reflection of well-being. *Journal of Personality, 65*(3).

Slawinska, M. M., & Davis, P. A. (2020). Recall of affective responses to exercise: Examining the influence of intensity and time. *Frontiers in Sports and Active Living, 2*, Article 573525. https://doi.org/10.3389/fspor.2020.573525

Sperandei, S., Vieira, M. C., & Reis, A. C. (2016). Adherence to physical activity in an unsupervised setting: Explanatory variables for high attrition rates among fitness center members. *Journal of Science and Medicine in Sport, 19*(11), 916–920. https://doi.org/10.1016/j.jsams.2015.12.522

Stevens, C. J., Baldwin, A. S., Bryan, A. D., Conner, M., Rhodes, R. E., & Williams, D. M. (2020). Affective determinants of physical activity: A conceptual framework and narrative review. *Frontiers in Psychology, 11*, 3366. http://doi:10.3389/fpsyg.2020.568331.

Strobach, T., Englert, C., Jekauc, D., & Pfeffer, I. (2020). Predicting adoption and maintenance of physical activity in the context of dual-process theories. *Performance Enhancement & Health, 8*(1), Article 100162. http://doi:10.1016/j.peh.2020.

Svebak, S., & Murgatroyd, S. (1985). Metamotivational dominance: A multimethod validation of reversal theory constructs. *Journal of Personality and Social Psychology, 48*(1), 107. https://doi.org/10.1037/0022-3514.48.1.107

Teixeira, D. S., Ekkekakis, P., Andrade, A., Rodrigues, F., Evmenenko, A., Faria, J., & D, M. (2021). Preference for and tolerance of the intensity of exercise questionnaire (PRETIE-Q): Validity, reliability and gender invariance in Portuguese health club exercisers. *Current Psychology, 1-14*. http://doi:10.1007/s12144-021-01718-3.

Teixeira, D. S., Rodrigues, F., Cid, L., & Monteiro, D. (2022). Enjoyment as a predictor of exercise habit, intention to continue exercising, and exercise frequency: The intensity traits discrepancy moderation role. *Frontiers in Psychology, 13*, 780059. https://doi.org/10.3389/fpsyg.2022.780059

Teques, P., Calmeiro, L., Silva, C., & Borrego, C. (2017). Validation and adaptation of the physical activity enjoyment scale (PACES) in fitness group exercisers. *Journal of Sport & Health Science, 9*(4), 352–357. http://doi:10.1016/j.jshs.2017.09.010.

Vazou-Ekkekakis, S., & Ekkekakis, P. (2009). Affective consequences of imposing the intensity of physical activity: Does the loss of perceived autonomy matter? *Hellenic Journal of Psychology, 6*(2), 125–144.

Weyland, S., Finne, E., Krell-Roesch, J., & Jekauc, D. (2020). How) does affect influence the formation of habits in exercise? *Frontiers in Psychology, 11*, Article 578108. https://doi.org/10.3389/fpsyg.2020.578108

Williams, D. M., Dunsiger, S., Ciccolo, J. T., Lewis, B. A., Albrecht, A. E., & Marcus, B. H. (2008). Acute affective response to a moderate-intensity exercise stimulus predicts physical activity participation 6 and 12 Months later. *Psychology of Sport and Exercise, 9*(3), 231–245. https://doi.org/10.1016/j.psychsport.2007.04.002

Williams, D. M., Dunsiger, S., Emerson, J. A., Gwaltney, C. J., Monti, P. M., & Miranda, R., Jr. (2016). Self-paced exercise, affective response, and exercise adherence: A preliminary investigation using ecological momentary assessment. *Journal of Sport & Exercise Psychology, 38*(3), 282–291. https://doi.org/10.1123/jsep.2015-0232

Williams, D. M., Dunsiger, S., Jennings, E. G., & Marcus, B. H. (2012). Does affective valence during and immediately following a 10-min walk predict concurrent and future physical activity? *Annals of Behavioral Medicine: A publication of the Society of Behavioral Medicine, 44*(1), 43–51. https://doi.org/10.1007/s12160-012-9362-9

Williams, D. M., & Evans, D. R. (2014). Current emotion research in health behavior science. *Emotion Review, 6*(3), 277–287. http://doi:10.1177/1754073914523052.

World Medical Association. (2013). World medical association declaration of Helsinki: Ethical principles for medical research involving human subjects. *JAMA, 310*(20), 2191–2194. https://doi.org/10.1001/jama.2013.281053

Zenko, Z., Ekkekakis, P., & Ariely, D. (2016). Can you have your vigorous exercise and enjoy it too? Ramping intensity down increases postexercise, remembered, and forecasted pleasure. *Journal of Sport & Exercise Psychology, 38*, 149–159. http://doi:10.1123/jsep.2015-0286.

Zenko, Z., O'Brien, J. D., Berman, C. J., & Ariely, D. (2017). Comparison of affect-regulated, self-regulated, and heart-rate regulated exercise prescriptions: Protocol for a randomized controlled trial. *Psychology of Sport and Exercise, 32*, 124–130. https://doi.org/10.1016/j.psychsport.2017.06.010